A Piecewise KL Bound from Hypergeometric Variance Envelopes

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Abstract

We integrate the pointwise minimum of two bounds on hypergeometric tilted variance: a finite-population quadratic envelope and the classical Poisson–binomial admissibility bound. The resulting rate has a closed piecewise form, with a scaled Bernoulli relative entropy in its central region and tangent-corrected binomial relative entropies in both tails. It strictly improves the binomial KL bounds obtained by sampling and success-count duality, and it improves the original scaled KL rate beyond two explicit switching points. We also compare it with the optimal uniform quadratic bound and prove that optimized factorial-moment bounds dominate MGF-based Chernoff bounds at integer thresholds.

1 Parameters and established variance inequalities

Throughout Sections 1–4, assume

$$N \geq 7, \qquad 3 \leq m \leq k \leq \lfloor N/2 \rfloor, \qquad N > 2m. \tag{1}$$

Both population parities are allowed. Let $H \sim \mathrm{Hyp}(N, k, m)$, and put

$$p = k/N, \quad q = 1 - p, \quad \mu = mp, \quad M = \frac{m(N-m)}{N-1}, \quad a = \frac{N-2m}{N-2}, \quad d = 1 - 2p.$$

All masses on $\{0, \ldots, m\}$ are positive. Write $\Lambda(t) = \log \mathbb{E}e^{tH}$, $u = \Lambda'(t)$, and $V(u) = \Lambda''(t(u))$, where $t(u)$ is the inverse tilted-mean map. The mean map is strictly increasing from $\mathbb{R}$ onto $(0, m)$.

We use the two inequalities established in the accompanying manuscript [1]:

$$V(u) \leq B(u) := \frac{u(m-u)}{m}, \tag{2}$$

$$V(u) \leq P(u) := Mpq + ad(u - \mu) - \frac{a^2}{M}(u - \mu)^2. \tag{3}$$

The first follows from the classical Jacobi real-root representation as a sum of independent Bernoulli variables; the second is its quadratic variance-envelope refinement. In (3), equality holds exactly at $u = \mu$. In particular, $P(\mu) = Mpq > 0$, and the endpoint calculations in [1] give $P(0) > 0$ and $P(m) > 0$. Concavity implies $P > 0$ on $[0, m]$. The results below are consequences and refinements of these established inequalities; their proof does not assert a new Jacobi factorization or a new two-level exchangeability reduction.

Define binary relative entropy by

$$D(z\|p) = z \log \frac{z}{p} + (1 - z) \log \frac{1 - z}{1 - p}, \qquad 0 \leq z \leq 1,$$

using the continuous endpoint values. All logarithms are natural.

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2 Integration in the tilted mean

Theorem 1 (Variance-envelope integration)**.** *Let X have finite support with minimum l, maximum h, and positive mass at both endpoints, with $l < h$. Let $\mu = \mathbb{E}X$ and $I(v) = \sup_t\{tv - \log \mathbb{E}e^{tX}\}$. Suppose its tilted variance $V(u)$ is bounded by a continuous function $Q(u) > 0$ on (l, h). Set*

$$J_Q(v) = \begin{cases} \displaystyle\int_\mu^v \frac{v-u}{Q(u)}\,du, & \mu \le v < h, \\ \displaystyle\int_v^\mu \frac{u-v}{Q(u)}\,du, & l < v \le \mu. \end{cases}$$

Then $I(v) \ge J_Q(v)$. Consequently

$$\Pr(X \ge v) \le e^{-J_Q(v)} \quad (v \ge \mu), \qquad \Pr(X \le v) \le e^{-J_Q(v)} \quad (v \le \mu).$$

The same statements hold at l, h whenever the displayed integrals have finite one-sided limits. If $V < Q$ on an interval of positive length between μ and v, then $I(v) > J_Q(v)$.

Proof. The tilted-mean map is strictly increasing, with derivative $V > 0$. For $v \in (l, h)$, differentiation of its Legendre transform gives $I'(v) = t(v)$, $I''(v) = 1/V(v)$, and $I(\mu) = I'(\mu) = 0$. Integrating twice gives exactly the displayed integrals with V in place of Q. Their kernels are nonnegative on the respective integration intervals, proving both the comparison and its strict version. Markov's inequality, optimized over $t \ge 0$ for the upper tail and $t \le 0$ for the lower tail, proves the tail assertions. At the endpoints, the finite-support exponential sum gives $I(l) = -\log \Pr(X = l)$ and $I(h) = -\log \Pr(X = h)$. Taking limits in the integral comparisons completes the proof. □

The rate obtained from P is

$$I_P(v) = \frac{M}{a^2} D\left(p + \frac{a}{M}(v-\mu) \,\middle\|\, p\right). \tag{4}$$

Indeed, $I_P'' = 1/P$ and $I_P(\mu) = I_P'(\mu) = 0$. Its KL argument remains strictly inside $(0, 1)$ for $0 \le v \le m$, since

$$1 - \frac{am}{M} = \frac{N(m-1)}{(N-2)(N-m)} > 0.$$

Thus an endpoint of the hypergeometric support never requires an invalid KL argument in (4). The corresponding rate for B is

$$I_B(v) = mD(v/m\|p). \tag{5}$$

For completeness, direct optimization of the established Laplace bound

$$\log \mathbb{E}e^{t(H-\mu)} \le \frac{M}{a^2}\{\log(q + pe^{at}) - pat\}$$

at threshold v gives $t_* = a^{-1}\log[\theta q/(p(1-\theta))]$, where $\theta = p + a(v-\mu)/M$, and exponent $I_P(v)$. This is the ordinary Bernoulli Chernoff transform. For an abstract Bernoulli envelope, $\theta = 0, 1$ are interpreted by $t \to \mp\infty$; values outside $[0, 1]$ are not inserted into D.

3 A closed piecewise rate

Put

$$c = \frac{1}{m} - \frac{a^2}{M}, \qquad h = (1-a)d, \qquad b = (m-M)pq,$$

and define

$$r_\pm = \mu + \frac{h \pm \sqrt{h^2 + 4cb}}{2c}. \tag{6}$$

Theorem 2 (Piecewise KL rate)**.** *Under* (1)*,* $0 < r_- < \mu < r_+ < m$*. The rate obtained by integrating* $1/\min(P, B)$ *is*

$$J(v) = \begin{cases} I_B(v) + I_P(r_-) - I_B(r_-) + (v - r_-)[I_P'(r_-) - I_B'(r_-)], & 0 \le v < r_-, \\ I_P(v), & r_- \le v \le r_+, \\ I_B(v) + I_P(r_+) - I_B(r_+) + (v - r_+)[I_P'(r_+) - I_B'(r_+)], & r_+ < v \le m. \end{cases} \tag{7}$$

It gives valid upper and lower tail bounds as in Theorem 1. For every $v \neq \mu$ *in* $[0, m]$*,*

$$I(v) > J(v) > I_B(v), \qquad J(v) \ge I_P(v), \tag{8}$$

and $J(v) > I_P(v)$ *holds exactly outside* $[r_-, r_+]$*.*

Proof. Writing $y = u - \mu$ gives $P(u) - B(u) = cy^2 - hy - b$. Here $b > 0$ and

$$c = \frac{(m-1)\{3N^2 - 4N - 4m(N-1)\}}{m(N-m)(N-2)^2} > 0,$$

because $m < N/2$ makes the bracket exceed $N(N-2) > 0$. The difference is negative at μ and positive at $0, m$; hence its two roots are precisely (6), with the asserted locations. It follows that $\min(P, B) = P$ between the roots and B outside. Integrating with zero value and derivative at μ yields (7); the affine corrections are forced by continuity of the rate and its first derivative at each root. In fact J is twice continuously differentiable at the roots because $P = B$ there.

Comparison of the nonnegative integration kernels gives $J \ge I_P, I_B$. On every noncentral path the interval near μ has $P < B$, proving $J > I_B$. A path leaving $[r_-, r_+]$ also crosses an interval where $B < P$, proving the characterization of $J > I_P$. Finally, $V < P$ except at μ, so the central interval gives $I > J$ on every noncentral path. At $0, m$, $1/B$ has simple poles, canceled by the endpoint integration kernels; the limits are finite and the same strict arguments apply. □

Corollary 3 (Integer and two-sided events)**.** *For real* $x > 0$*, set* $z_+ = \lceil \mu + x \rceil$ *and* $z_- = \lfloor \mu - x \rfloor$*. Use* $e^{-J(z_+)}$ *for the upper event when* $z_+ \le m$*, and zero otherwise; use* $e^{-J(z_-)}$ *for the lower event when* $z_- \ge 0$*, and zero otherwise. Their sum, capped at one, bounds* $\Pr(|H - \mu| \ge x)$*.*

4 Comparison boundaries

4.1 Four binomial KL bounds

Sampling duality and complementation give the following four classical binomial Chernoff rates at a fixed threshold v:

$$R_1(v) = mD(v/m \| k/N), \qquad R_2(v) = kD(v/k \| m/N),$$
$$R_3(v) = (N-m)D((k-v)/(N-m) \| k/N), \qquad R_4(v) = (N-k)D((m-v)/(N-k) \| m/N).$$

All have value and derivative zero at μ. Their reciprocal second derivatives are, respectively,

$$B_1 = B, \quad B_2 = \frac{v(k-v)}{k}, \quad B_3 = \frac{(k-v)(N-m-k+v)}{N-m}, \quad B_4 = \frac{(m-v)(N-k-m+v)}{N-k}.$$

With $L = k - m \geq 0$ and $R = N - m - k \geq 0$,

$$\begin{aligned} B_2 - B &= \frac{(k-m)v^2}{mk} \geq 0, \\ B_4 - B &= \frac{(N-k-m)(m-v)^2}{m(N-k)} \geq 0, \\ B_3 - B &= \frac{Lv^2 + R(m-v)^2 + mLR}{m(N-m)} \geq 0. \end{aligned}$$

Therefore $R_1 = I_B \geq R_j$ for $j = 2, 3, 4$ on $[0, m]$, and Theorem 2 strictly improves their maximum at every noncentral threshold. Duality itself is classical and is explicitly used by Mannalath, Zapatero and Curty [4]; the improvement here comes from integrating the smaller mean-dependent variance bound.

4.2 The optimal uniform quadratic coefficient

Define

$$\kappa_N = \begin{cases} 1, & N \text{ even}, \\ [N \operatorname{artanh}(1/N)]^{-1}, & N \text{ odd}, \end{cases} \qquad V_* = M\kappa_N/4.$$

This is the coefficient for $H - \mu$ obtained from the exact uniform exchangeability constant in [1], in the convention $\log \mathbb{E}e^{t(H-\mu)} \leq V_* t^2/2$. The optimal Bernoulli proxy is

$$\beta(p) = \frac{1-2p}{2\log((1-p)/p)}, \qquad \beta(1/2) = 1/4.$$

Under (1), $M\beta(p) \leq V_*$. Consequently the whole scaled Laplace envelope is bounded above by $M\beta(p)t^2/2$, and conjugation gives

$$J(v) \geq I_P(v) \geq \frac{(v-\mu)^2}{2M\beta(p)} \geq \frac{(v-\mu)^2}{2V_*}. \tag{9}$$

Thus the scaled KL and piecewise bounds are never weaker than this particular uniform quadratic bound. They need not dominate the numerically optimized quadratic proxy for each fixed hypergeometric law.

For $p < 1/2$, equality in the middle inequality at a nonzero displacement requires $\theta = 1 - p$, hence $v - \mu = M(1-2p)/a > 0$; this follows from the unique nonzero equality in the Bernoulli proxy inequality. For odd N, equality with the final uniform rate additionally requires $k = (N-1)/2$. For even N, equality $M\beta(p) = V_*$ requires $p = 1/2$, but the balanced Bernoulli quadratic inequality is strict at nonzero tilt. The inequality $J \geq I_P$ is an equality precisely in the central region.

Writing $x = v - \mu$, the local expansion is

$$I_P(\mu + x) = \frac{x^2}{2Mpq} - \frac{a(1-2p)x^3}{6M^2p^2q^2} + \frac{a^2(p^3+q^3)x^4}{12M^3p^3q^3} + O(x^5).$$

This shows the directional asymmetry and the variance-scale improvement near the mean. Since $M < m$, $I_P > I_B$ locally, but the unspliced I_P can be smaller than I_B in the tails. Theorem 2 removes this loss relative to the four binomial bounds.

4.3 Exact-variance Bennett and Bernstein bounds

Put $v_0 = Mpq$ and $h_0(z) = (1+z)\log(1+z) - z$. The Bennett exponent in Greene–Wellner [5, Theorem 3] is $v_0 h_0(x/v_0)$; their Bernstein corollary replaces it by $x^2/[2(v_0 + x/3)]$. For either tail, the scaled rate is strictly larger than the Bennett exponent at every nonzero supported deviation. Indeed, for $s > 0$,

$$P(\mu + s) = v_0 + ads - a^2 s^2/M < v_0 + s, \qquad P(\mu - s) = v_0 - ads - a^2 s^2/M < v_0 + s,$$

because $0 \le ad < 1$. The second derivative of $v_0 h_0(x/v_0)$ is $1/(v_0 + x)$, while that of $I_P(\mu \pm x)$ is $1/P(\mu \pm x)$. Both rates and their first derivatives vanish at zero. Integrating the strict comparison twice, and then using $J \ge I_P$, proves the assertion. The standard inequality $h_0(z) \ge z^2/[2(1+z/3)]$ gives the Bernstein comparison as well. This does not compare J with Greene–Wellner's other bounds containing a prefactor and a quartic exponent.

4.4 Optimized factorial moments

Proposition 4 (A comparison with exact Chernoff)**.** *Let X be nonnegative and integer-valued with bounded support, and let $d \ge 1$ be an integer. Define*

$$F_d = \min_{0 \le j \le d} \frac{\mathbb{E}\binom{X}{j}}{\binom{d}{j}}.$$

Then

$$\Pr(X \ge d) \le F_d \le \inf_{t \ge 0} \mathbb{E}e^{t(X-d)}.$$

For the hypergeometric law,

$$\frac{\mathbb{E}\binom{H}{j}}{\binom{d}{j}} = \frac{(k)_{\underline{j}}(m)_{\underline{j}}}{(N)_{\underline{j}}(d)_{\underline{j}}}, \qquad 0 \le j \le d.$$

In particular, at integer upper thresholds $d > \mu$ in (1)*,*

$$\Pr(H \ge d) \le F_d \le e^{-I(d)} < e^{-J(d)}.$$

The lower-tail version follows by applying the result to $m - H$.

Proof. For $j \le d$, $\binom{X}{j} \ge \binom{d}{j}$ on $\{X \ge d\}$, which proves the first inequality. For $z = e^t - 1 \ge 0$, the binomial theorem gives

$$\mathbb{E}(1+z)^X = \sum_{j \ge 0} \mathbb{E}\binom{X}{j} z^j \ge F_d \sum_{j=0}^{d} \binom{d}{j} z^j = F_d (1+z)^d.$$

Divide by e^{td} and take the infimum. Hypergeometric factorial moments give the displayed ratio. □

Optimized hypergeometric factorial-moment bounds are already studied in [4]; neither that method nor its optimization is new here. For $\mu < d \le \min(k, m)$, a valid minimizing index is

$$j_* = \left\lceil \frac{Nd - km}{N + d - k - m} \right\rceil \in \{1, \dots, d\}.$$

Indeed the ratio of consecutive candidates is at most one exactly when $j \le (Nd - km)/(N + d - k - m)$. When this fraction is integral, the two adjacent indices tie if both are admissible. At $d = \min(k, m)$ the valid endpoint index is $j = d$; one must not use the undefined denominator $(d)_{\underline{d+1}}$. These facts set a precise comparison boundary: the piecewise KL expression is an explicit refinement of the stated KL and uniform quadratic envelopes, not a universal improvement over optimized factorial moments or all existing finite-population bounds.

References

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